# Yaoxuan (Thaddy) Wu

## Education

**University of California, Los Angeles** — **Los Angeles, CA**
*Ph.D. Student* — *Sep. 2022 – Present*

**University of California, Los Angeles** — **Los Angeles, CA**
*M.S.* — *Sep. 2022 – Mar. 2025*

**Peking University** — **Beijing, China**
*B.S., Turing Class, Summa Cum Laude* — *Sep. 2018 – Jun. 2022*

## Internship

**Amazon Web Services** — **Arlington, VA**
*Research Scientist Intern* — *Jun. 2025 – Sep. 2025*

## Research Overview

- **AI Systems.** I develop AI techniques for software engineering by combining large language models with symbolic reasoning and property-based methods. My work spans LLM-based test generation, automated proof synthesis, and cloud system simulation, and is expanding toward AI-native software systems and autonomous software agents. Representative projects include *PALM* (*ICPC 2026*), *DualVeri* (Preprint, 2026), and *CloudTwin* (Preprint, 2026).
- **Testing and Verification for DISC Systems.** I develop testing and verification techniques for Apache Spark and other data-intensive scalable computing (DISC) systems using symbolic execution, fuzzing, and property-based testing. Representative projects include *NaturalSym* (*FSE 2024*), *NaturalFuzz* (*ASE 2023*), *DISCPBT* (Preprint, 2026), and the position paper *Why Property-Based Testing* (*FSE DISE Workshop 2026*).

## Publications

* Equal contribution.

**Preprint** The Parser Already Knows: Lightweight Bias Correction in Constrained Decoding.
Isil Özgü, **Yaoxuan Wu**, Guy Van den Broeck, and Miryung Kim. arXiv:2608.10137 (2026).

**Preprint** DualVeri: Agentic Proof and Property-Based Testing via Property-Templates in Data-Intensive Computing.
Seongmin Lee*, **Yaoxuan Wu***, and Miryung Kim. arXiv:2607.09072 (2026).

**Preprint** DISCPBT: Operationalizing Property-Based Testing for Data-Intensive Scalable Computing Systems.
**Yaoxuan Wu**, Ingrid Lee, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim. arXiv:2606.11132 (2026).

**Preprint** CloudTwin: Inverted Tool Use for Long-Horizon Cloud Simulation.
Yifan Zhang*, **Yaoxuan Wu***, Leyi Cui*, Hui Guan, Brandon Paulsen, Joey Dodds, and Daniel Kroening.

**Preprint** DAG-based Workload Generation for Testing Data-Intensive Scalable Computing Frameworks.
Ahmad Humayun, **Yaoxuan Wu**, Muhammad Ali Gulzar, and Miryung Kim.

**FSE'26 WS** Why Property-Based Testing is Necessary for Data-Intensive Scalable Computing.
**Yaoxuan Wu**, Ingrid Lee, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim.

**ICPC'26** PALM: Path-aware LLM-based Test Generation with Comprehension.
**Yaoxuan Wu**, Xiaojie Zhou, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim.

**Preprint** Targeted Testing of Compiler Optimizations via Grammar-Level Composition Styles.
Zitong Zhou, Ben Limpanukorn, Hong Jin Kang, Jiyuan Wang, **Yaoxuan Wu**, Ákos Kiss, Renata Hodovan, and Miryung Kim. arXiv:2512.04344 (2025).

**FSE'24** Natural Symbolic Execution-based Testing for Big Data Analytics.
**Yaoxuan Wu**, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim.

**ASE'23** NaturalFuzz: Natural Input Generation for Big Data Analytics.
Ahmad Humayun, **Yaoxuan Wu**, Miryung Kim, and Muhammad Ali Gulzar.

**USENIX'23** Pelican: Exploiting Backdoors of Naturally Trained Deep Learning Models in Binary Code Analysis.
Zhuo Zhang, Guanhong Tao, Guangyu Shen, Shengwei An, Qiuling Xu, Yingqi Liu, Yapeng Ye, **Yaoxuan Wu**, and Xiangyu Zhang.

## Awards

**2024-** Amazon Fellowship
**2023-** UCLA Graduate Summer Research Scholarship
**2022-** UCLA Computer Science Department Scholar Award
**2021-** Jiukun Scholarship
**2020-** John Hopcroft Scholarship
**2018-2021** ACM-ICPC Asia Regional and EC Final Gold/ Silver Medals
**2017** National Olympiad in Informatics (China), Gold Medal

**Current address**: Los Angeles, California, USA
**Citizenship**: China